رواد مصر الرقمية

وزارة الاتصالات
وتكنولوجيا المعلومات

Telecom Churn

Why customers leave!

Behind the project

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Introduction

Imagine a customer who relies on their telecom service every day making calls, paying bills, staying connected.
Then one day, they decide to leave for a competitor !
Every lost customer means lost revenue and higher costs to win new ones.

The challenge

High customer churn in the telecom sector causes major revenue losses and raises the cost of acquiring new customers.

- Strong competition
- Service quality issues
- Pricing and contracts
- Customer support problems
- At-risk customers

Objectives

1. **Identify churn drivers:** analyze customer characteristics, contracts, and service usage to reveal why customers leave.

2. **Support decision-making:** build an interactive dashboard to track churn trends and guide retention strategies.

3. **Quantify business impact:** measure revenue loss caused by churn and highlight the most affected segments.

Key Questions

1. What is the churn rate and how does it change over time?
2. What are the main reasons behind churn ?
3. What is the financial cost and who are the high-value customers to retain ?
4. Which customer segments are most at risk of leaving ?
5. How do pricing, contracts, and services influence churn ?

About Dataset

This dataset includes telecom customer information such as demographics, service subscriptions, contract types, usage patterns, charges, and churn status. It covers a large number of customers over recent periods.

Methodology & Data preparation

Data Gathering
Collect telecom customer churn data.

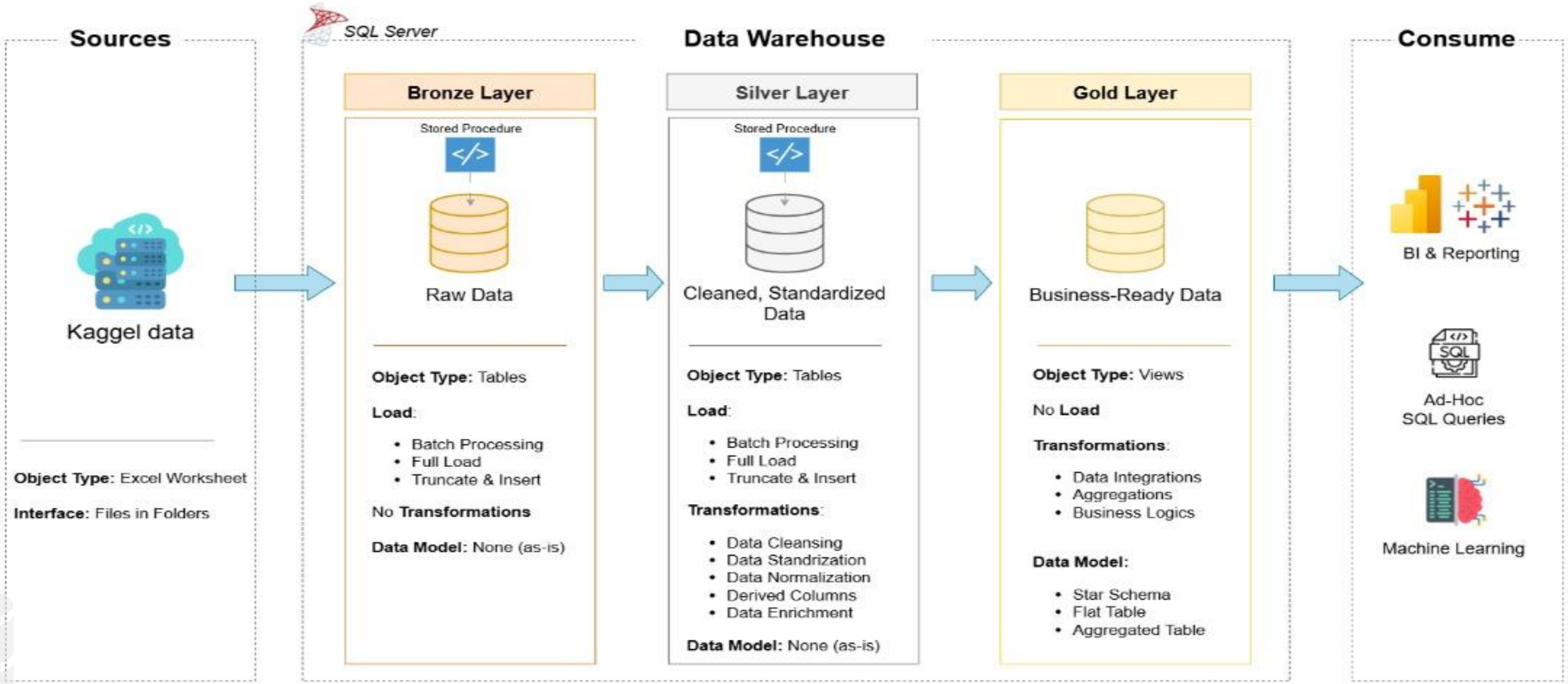
Cleaning & prep
Fix errors, using data warehouse (silver/bronze/gold)

Modeling & linking
Build relationships, structured datasets.

Analysis & KPIs
Calculate metrics, extract insights.

Visualization
Create interactive dashboards & visuals.

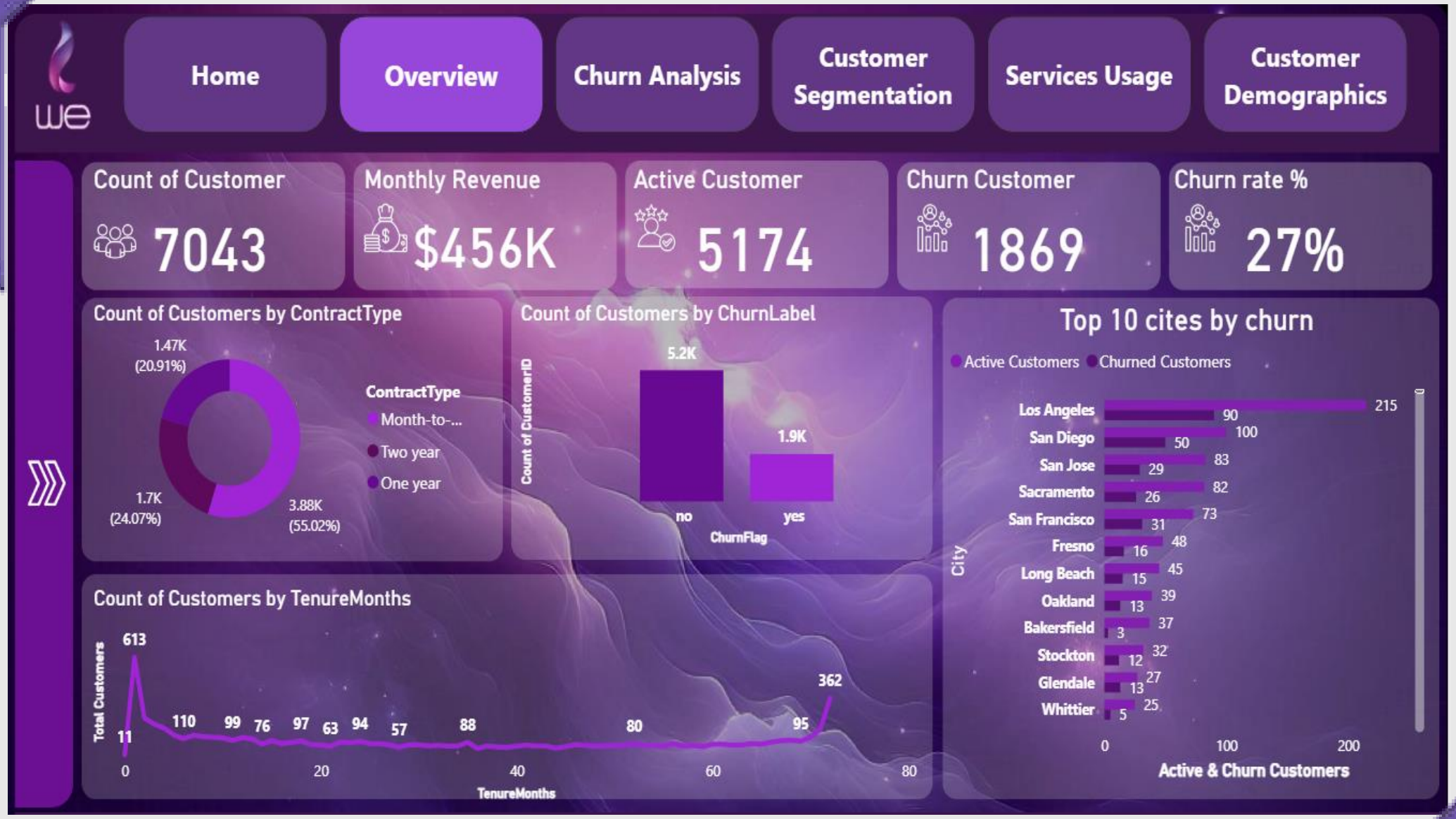
High Level Architecture



Dashboards Overview

Power BI Dashboards Analysis delivers interactive insights for the Telecom Churn project, covering Overview, Churn Analysis, Customer Segmentation, Service Usage, and Customer Demographics, enabling stakeholders to make informed, data-driven decisions.

Overview


we
Home
Overview
Churn Analysis
Customer Segmentation
Services Usage
Customer Demographics
Count of Customer
7043
Monthly Revenue
$456K
Active Customer
5174
Churn Customer
1869
Churn rate %
27%
Count of Customers by ContractType
1.47K (20.91%)
1.7K (24.07%)
3.88K (55.02%)
ContractType
Month-to-...
Two year
One year
Count of Customers by ChurnLabel
Count of CustomerID
5.2K
1.9K
no
yes
ChurnFlag
Top 10 cites by churn
Active Customers
Churned Customers
City
Los Angeles 215 90
San Diego 100 50
San Jose 83 29
Sacramento 82 26
San Francisco 73 31
Fresno 48 16
Long Beach 45 15
Oakland 39 13
Bakersfield 37 3
Stockton 32 12
Glendale 27 13
Whittier 25 5
0
100
200
Active & Churn Customers
Count of Customers by TenureMonths
Total Customers
613
11
110
99
76
97
63
94
57
88
80
95
362
0
20
40
60
80
TenureMonths

Insights

- Indicating a significant portion of users leaving the service.

- Most customers are on month-to-month contracts (55%), showing flexibility but also higher churn risk.

Insights

- Majority of users have shorter tenure months, meaning early-stage customer loss is high.

- Top churn cities (Los Angeles, San Diego) reflect regional retention challenges and competitive market pressure.

Churn Analysis

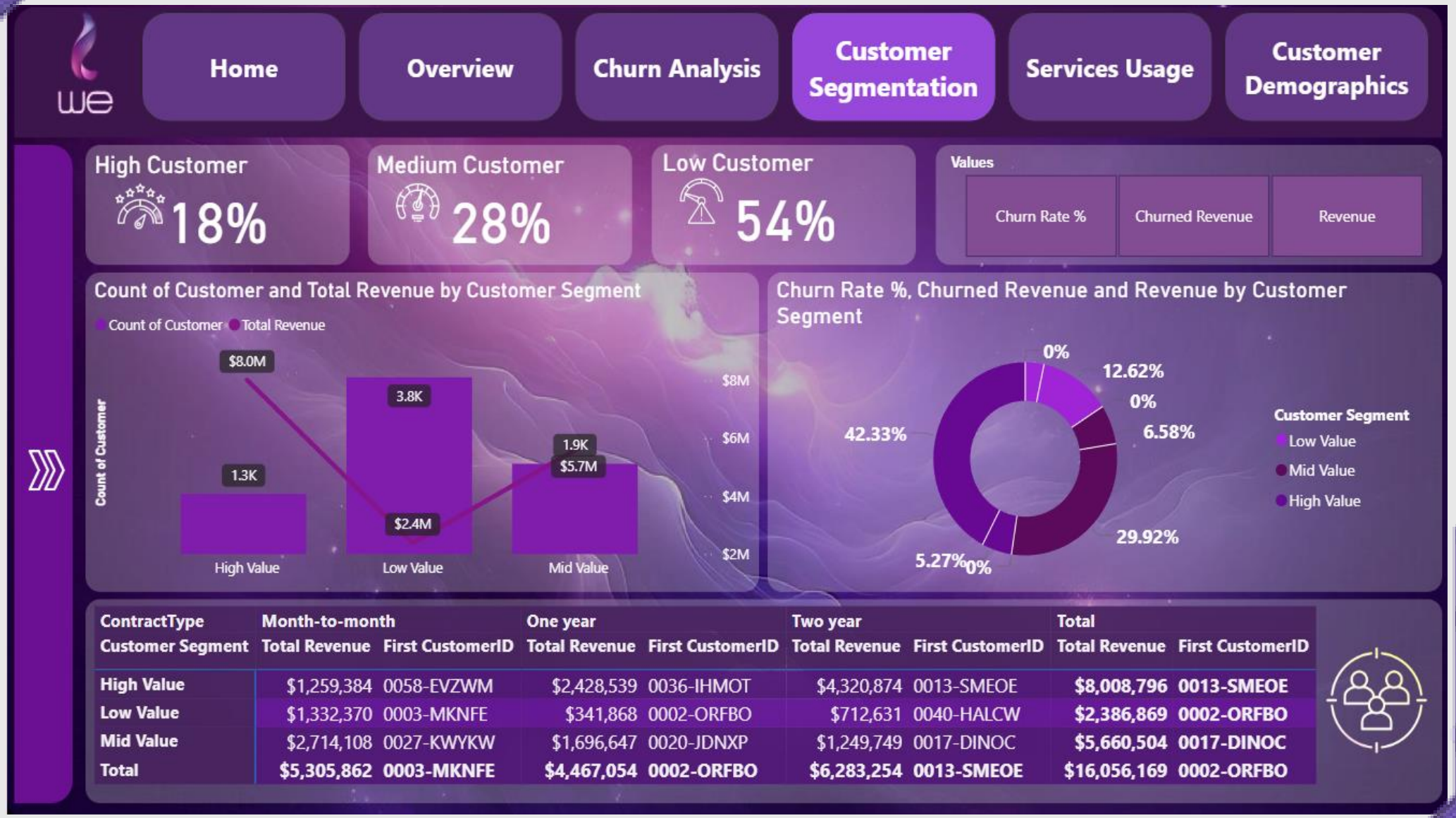
Insights

Average customer tenure is 32 months, meaning customers usually stay for some time before deciding to leave.

Insights

- Many churned customers use electronic check payments, which may reflect lower digital engagement or convenience issues.
- Male customers and non-senior citizens show slightly higher churn tendencies.
- Most churned customers are on month-to-month contracts, short-term contracts make it easier for customers to leave.

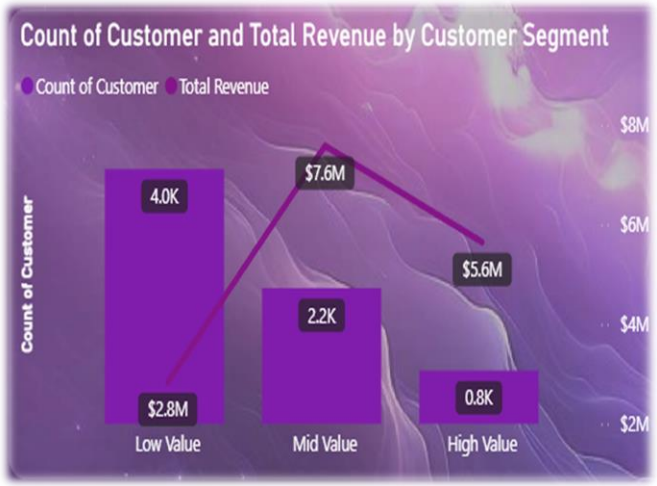
Customer Segmentation

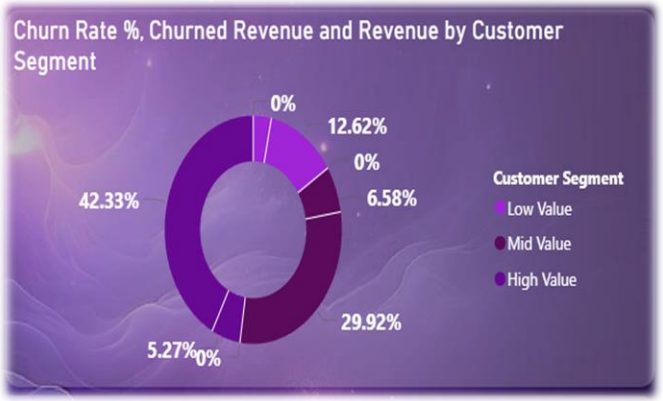


ContractType	Month-to-month		One year		Two year		Total	
Customer Segment	Total Revenue	First CustomerID	Total Revenue	First CustomerID	Total Revenue	First CustomerID	Total Revenue	First CustomerID
High Value	$1,259,384	0058-EVZWM	$2,428,539	0036-IHMOT	$4,320,874	0013-SMEOE	$8,008,796	0013-SMEOE
Low Value	$1,332,370	0003-MKNFE	$341,868	0002-ORFBO	$712,631	0040-HALCW	$2,386,869	0002-ORFBO
Mid Value	$2,714,108	0027-KWYKW	$1,696,647	0020-JDNXP	$1,249,749	0017-DINOC	$5,660,504	0017-DINOC
Total	$5,305,862	0003-MKNFE	$4,467,054	0002-ORFBO	$6,283,254	0013-SMEOE	$16,056,169	0002-ORFBO

Insights

- The Low Value segment forms the majority (57%) but contributes only $2.8M, showing low profitability.
- Mid and High Value customers together (43%) generate most of the revenue, $7.6M and $5.6M respectively, making them the company's most valuable segments.
- Churn Rate is highest among Mid Value (42.33%), followed by Low Value (29.82%), and lowest for High Value (12.62%), indicating that higher-value customers are more loyal.





Insights

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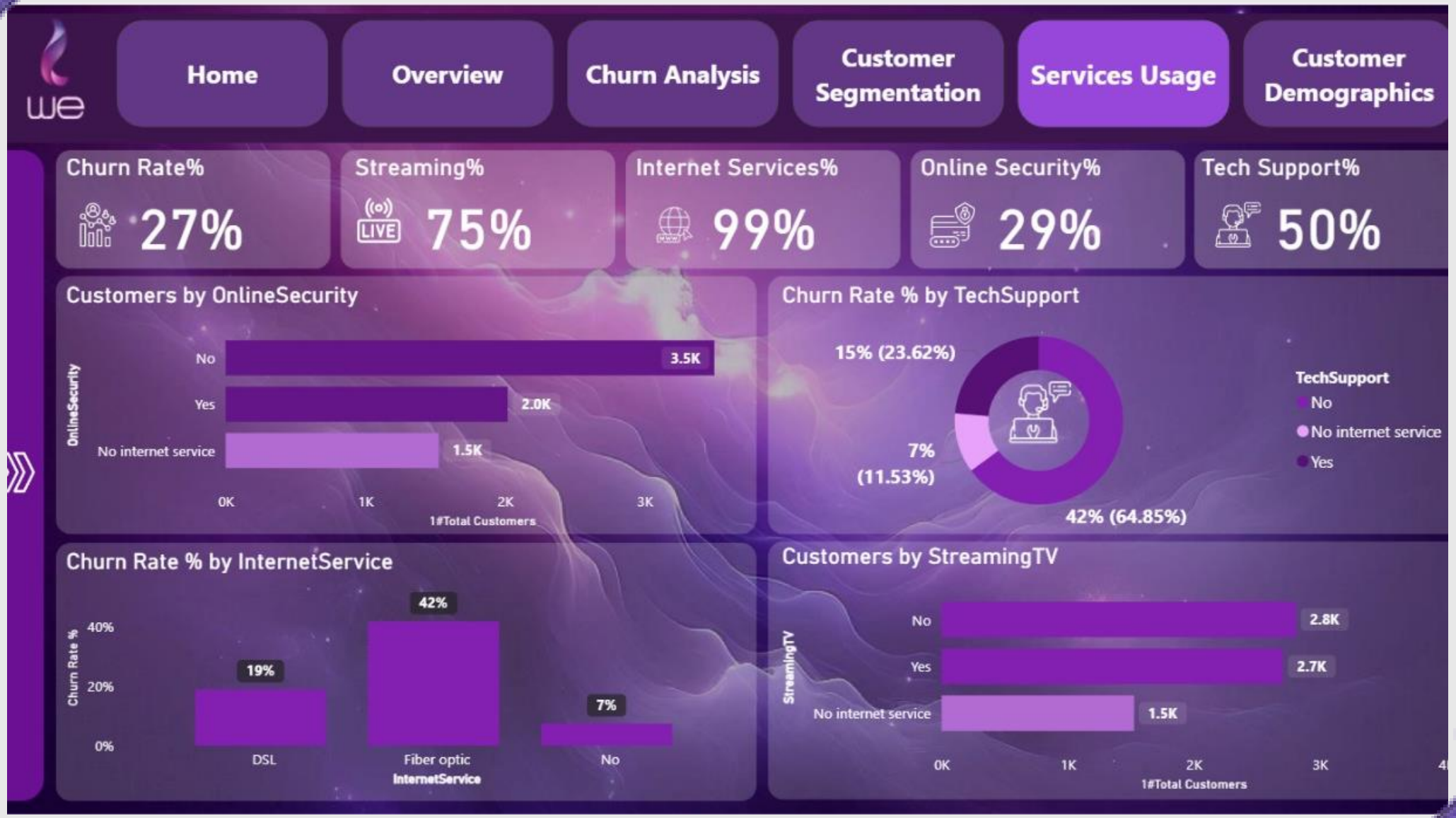
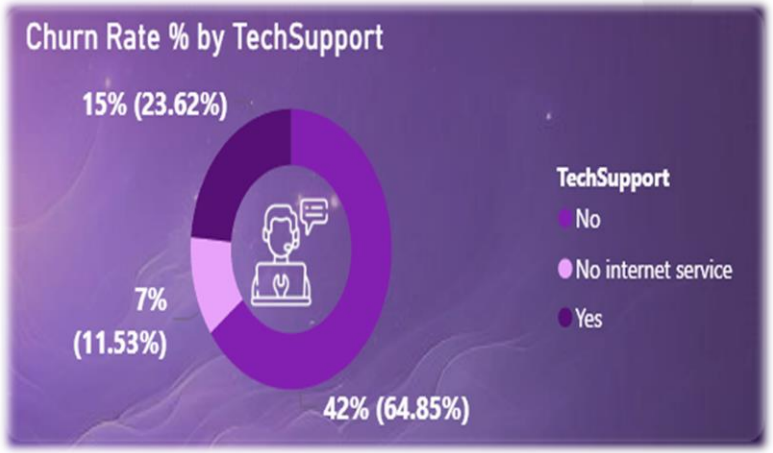
- Most churned customers have month-to-month contracts, especially in the Low Value segment (2,760 customers).
- Two-year contracts show the lowest churn and highest revenue, confirming that longer commitments reduce customer loss.
- The total revenue across all segments equals $16M, consistent with overall churn analysis.

Services Usage

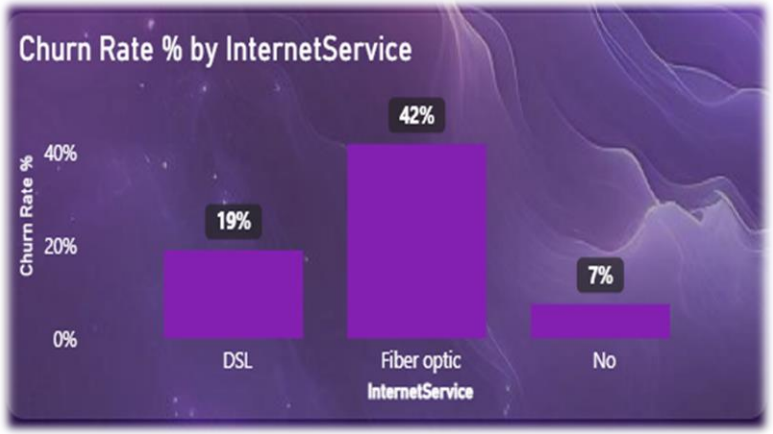


Insights

- Customers who utilize Tech Support exhibit a high churn rate of 42%, suggesting that interacting with support is a major pre-cursor or consequence of a poor overall experience.

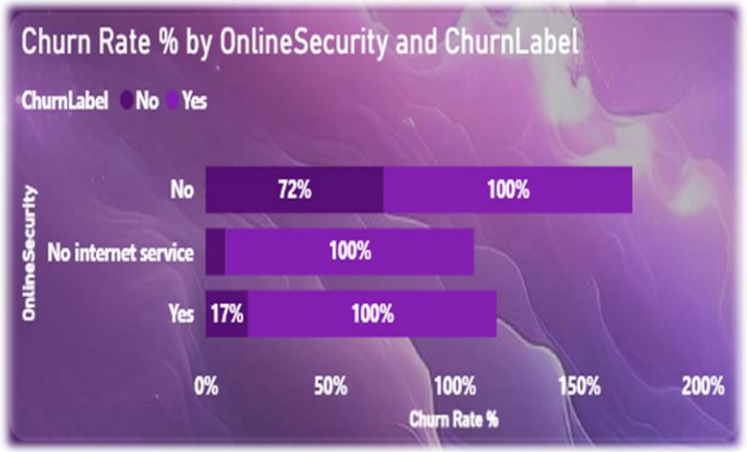


- The Fiber Optic internet type has the highest churn rate at a critical 42%, significantly surpassing DSL's 19% churn and pointing to systemic issues with this specific product line.

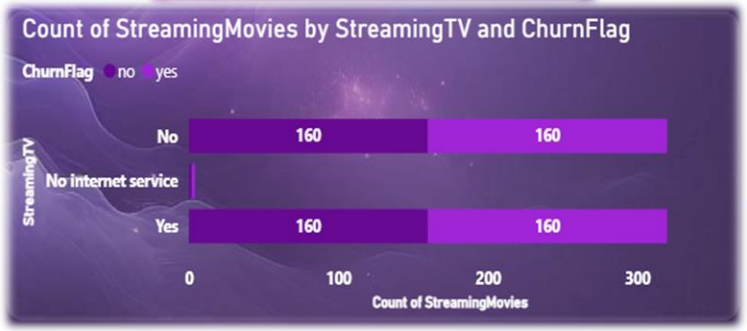


Insights

- This service is the strongest safeguard against churn, drastically reducing the churn rate from 72% (for those without it) down to just 17% for those who subscribe.



- Internet service is the most widely adopted product (78% penetration), while the volume of Streaming Movies Watched is not a differentiator between churned and non-churned customers.

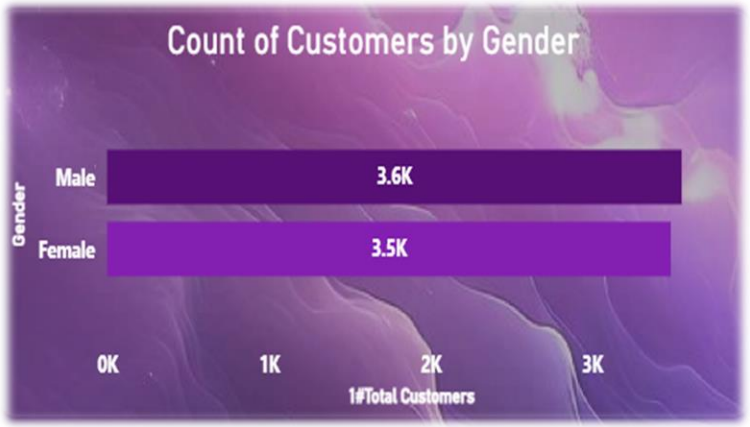


Customer Demographics

Insights

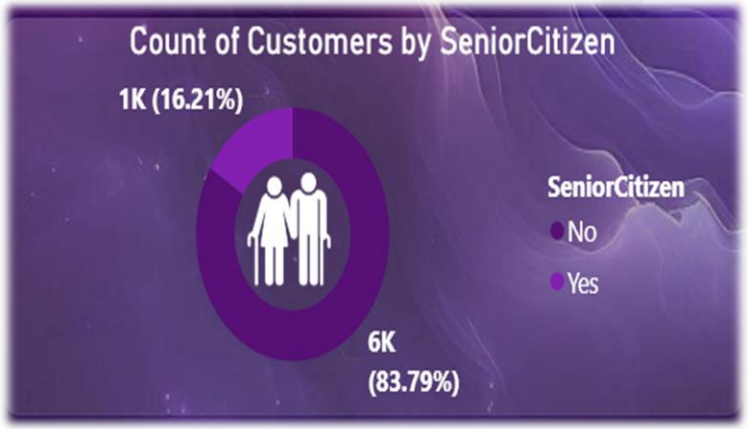
34% of customers do not have a Partner; this large single-customer segment represents the most volatile group in terms of churn risk, despite being smaller than the Partner segment.

Gender is the least relevant demographic factor for churn, as the customer count for Male (3.6K) is almost identical to Female (3.5K), showing no correlation with overall churn behavior.

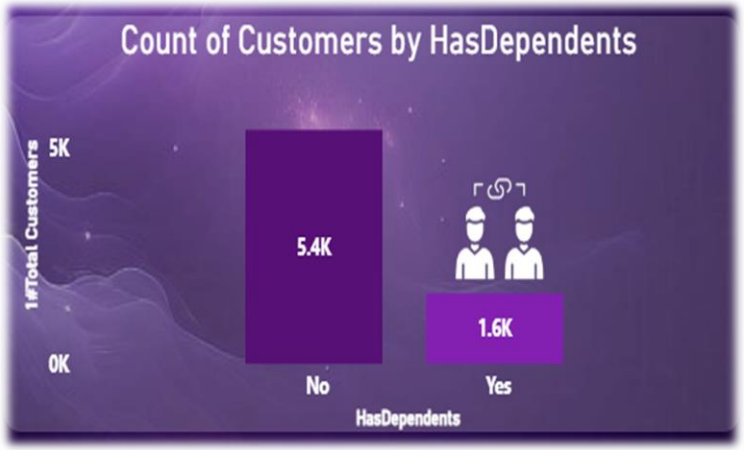


Insights

- The Senior Citizen segment (16.21%) is the second smallest group; while they are stable, they likely have the lowest revenue impact due to low Average CLTV and volume.



- The group with Dependents (1.6K) is the smallest segment, this group is generally stable, its size means its low churn rate has a minimal positive impact on the overall retention rate.



Recommended

VALUE & CONTRACT STABILITY:

- **Reward High Value** with **Premium Support**.
- **Promote Long-Term Contracts** using **Incentives**.
- **Target Low/Mid Value** with **Personalized Programs**.

URGENT PRODUCT QUALITY FIXES

- **Fix Fiber Optic Network** issues causing **42% churn**.
- **Mandate Online Security Bundling** to stabilize the Fiber base.

IMPROVE CUSTOMER EXPERIENCE:

- **Redesign Tech Support Process** to improve **First-Contact Resolution (FCR)**.

Link

Telecom churn analysis